UK Data Dashboard Project

UK Local Authority Greenhouse Gas Emissions — Interactive Excel Dashboard

SkillVersed Data Analytics | Excel Level 2
Prepared by Shraddha Yadav

This report explains how I built my interactive Excel dashboard for the UK Data Dashboard Project. It follows the five stages set out in the brief: choosing a dataset, evaluating and documenting it, identifying who the dashboard is for, deciding on the perspective and the analysis, and finally designing and building the dashboard itself. I have tried to explain my thinking at each step and the choices I made along the way.

1. Dataset Selection

For this project I chose a UK government dataset on greenhouse gas emissions. The full title is “UK local authority and regional greenhouse gas emissions statistics, 2005 to 2023”, and it is published by the Department for Energy Security and Net Zero (DESNZ) on GOV.UK. It is released as official national statistics under the Open Government Licence, so it is free to use as long as the source is credited.

I picked this dataset for a few reasons. First, it is a real public-interest topic. Emissions and the journey to net zero are talked about a lot at the moment, and councils across the UK have to report and reduce their emissions, so the data actually gets used by real people. Second, it is a proper UK-wide dataset rather than information about a single company or a single product, which is what the brief asked for. Third, it has a good mix of things to work with: places (countries, regions and local authorities), a long run of years, different categories (sectors, sub-sectors and gases) and several numeric measures. That mix is exactly what makes a dataset suitable for a dashboard, because it gives me different ways to slice and compare the numbers.

The original file from GOV.UK is very large (around 79 MB), which is far more than I need for this assignment and would make the workbook slow. So I took a working extract of exactly 3,000 records that still keeps the full structure of the data. I kept the original source link in the workbook so anyone can trace it back to the official release.

2. Dataset Evaluation and Documentation

Before building anything, I looked through the data to understand what each column meant and to check that it was good enough to use. The extract has 3,000 rows and 19 columns. Each row is one combination of a local authority, a sector and sub-sector, a greenhouse gas and a year, with the emissions figure and some supporting information. The main groups of columns are below.

Group	Columns	What it tells me
Geography	Country, Region, Local Authority (plus ONS codes)	Where the emissions happen, from country level down to individual councils.

Group	Columns	What it tells me
Time	Calendar Year (2005–2023)	Lets me look at the trend over time.
Category	LA GHG Sector, Sub-sector, Greenhouse gas	What kind of activity and which gas the emissions come from.
Measures	Territorial emissions (kt CO2e), Scope-of-influence emissions (kt CO2), Population, Area, Emissions per capita, Emissions density	The actual numbers I can total, average and compare.
Banding	Emissions band	A simple grouping of the emissions size, useful for counting records.

I then ran a few simple quality checks. I confirmed there were no blank rows and that each record had a country, region, year, sector and gas. I checked that the year values all sat between 2005 and 2023, that the gases were limited to CO2, CH4 and N2O, and that the sectors matched the eight categories the government uses (Industry, Domestic, Transport, Public Sector, Commercial, Agriculture, Waste and LULUCF). I also checked that the numeric columns really were numbers so that my totals and averages would work. The per-capita and density columns are worked-out measures (emissions divided by population or area), which are handy for fair comparisons between large and small areas.

To document the dataset properly, the workbook has two supporting sheets. The Source sheet records where the data came from, the publisher, the licence, the time period and the size of my extract, with the official links. The Data Dictionary sheet lists every one of the 19 columns with its type and a plain-English description, so anyone opening the file can understand each field without having to ask me. Overall I found the dataset to be reliable and well structured. Its main limitation is simply that I am using a sample of 3,000 rows rather than the entire national file, so the totals reflect my extract rather than the whole country, and I have been clear about that throughout.

3. End-Consumer Identification

I designed the dashboard for a non-technical audience rather than for data analysts. The people I had in mind are council officers and sustainability or policy teams who need to see how emissions break down for their area, local councillors and decision-makers who want the headline numbers quickly, and interested members of the public or students looking at where UK emissions come from.

These users are comfortable reading charts and big numbers, but they should not have to write formulas, build pivot tables or understand the raw data to get value from the dashboard. That shaped a lot of my design choices. The questions I imagined them asking were things like:

- Which regions and sectors produce the most emissions?
- How have emissions changed between 2005 and 2023?
- Which greenhouse gas makes up most of the total?

- How does my own country or region compare with the rest of the UK?

Because of this audience, I kept the language simple, put the most important numbers at the top, and made the filtering as easy as clicking a button rather than typing anything. The aim is that someone can open the file and start exploring within a few seconds.

4. Perspective and Analysis Definition

The perspective I took is: “Where do the UK’s greenhouse gas emissions come from, and how have they changed?” In other words, I wanted the dashboard to break the national total down by place, by sector, by gas and over time, and to let the user compare those breakdowns for themselves.

To support that story, I decided on six headline figures (KPIs) that sit across the top of the dashboard. I chose these because they summarise the dataset in a way the audience cares about, and they all update as the user filters.

KPI	What it shows
Total emissions (kt CO2e)	The total territorial emissions for whatever the user has selected.
Records in view	How many rows of the dataset are currently included.
Scope emissions (kt CO2)	The total scope-of-influence emissions for the selection.
CO_2 emissions (kt CO2e)	How much of the selected total is carbon dioxide.
Average emissions per record	The average emission value per row, as a sense of typical size.
CO_2 share of total	The percentage of emissions made up by carbon dioxide.

For the analysis itself, I decided to let the user filter the whole dashboard on five things: Country, Region, Calendar Year, Sector and Greenhouse gas. These are the five dimensions that the audience is most likely to want to focus on, and together they let someone answer almost any “what about…?” question, for example “what about transport in Scotland” or “what about methane in 2023”. Every chart and every KPI responds to these filters at the same time, so the whole page always tells one consistent story.

5. Dashboard Design and Development

I built the dashboard on its own sheet so it reads like a finished product, with the supporting data, pivot tables, dictionary and source kept on separate sheets behind it. The layout works from top to bottom: a clear title and source line, then the six KPI cards, then the filter controls, and then the charts.

Interactivity — slicers

The most important feature is that the dashboard is genuinely interactive. I used five slicers — Country, Region, Calendar Year, Sector and Greenhouse gas — which are the clickable filter buttons in Excel. The slicers are connected to the pivot tables that sit behind every chart, so when the user clicks a value, all eight charts and all six KPIs filter together at once. I chose

slicers rather than ordinary drop-down cells because they are much friendlier for a non-technical user: you can see all the options laid out, click more than one, and clear them again easily, without needing to know anything about how the workbook is put together.

Charts

I included eight charts and deliberately used a few different chart types so each piece of information is shown in the way that suits it best:

- Territorial emissions by region — a bar chart, good for comparing 13 regions.
- Emissions by sector — a column chart, to compare the eight sectors.
- Emissions trend by year — a line chart, the natural choice for change over time.
- Emissions by greenhouse gas — a doughnut chart, to show the share of the three gases.
- Emissions by country — a bar chart, comparing the UK nations.
- Records by emissions band — a column chart, showing how the records are distributed by size.
- Scope-of-influence emissions by region — a bar chart, sitting alongside the territorial figure for context.
- Average emissions per capita by country — a bar chart, giving a fairer per-person comparison.

How it is built underneath

Behind the dashboard I created eight pivot tables, one for each chart, all drawing from the same dataset. The charts read from those pivot tables, and the KPI cards use GETPIVOTDATA formulas that read the pivot totals. This is what makes the slicers control everything: a slicer filters the pivot tables, the pivot totals change, and the charts and KPIs follow automatically. I wrapped the KPI formulas so that they show zero rather than an error if a filter ends up with no data, which keeps the dashboard looking tidy whatever the user clicks.

Design choices for clarity

I kept the visual style simple and consistent so it is easy on the eye and works for a non-technical reader. I used a calm colour scheme, gave every chart a clear title with its units, and used thousands separators on the large emission numbers so they are readable. The KPI cards use large, bold figures so the headline story is obvious at a glance. I avoided clutter and decorative effects, because for this audience clarity matters more than decoration.

What the dashboard shows

With everything in place, a few clear findings stand out. Across my 3,000-record extract the total territorial emissions come to about 59,538 kt CO2e. Carbon dioxide is by far the largest part of the mix at roughly 86 per cent, with methane and nitrous oxide making up the rest. The North East shows the highest regional total and Domestic is the single largest sector. The user can confirm any of this for themselves, or look at their own area, simply by clicking the slicers.

Finally, I tested the dashboard by clicking through different combinations of slicers to make sure the charts and KPIs all updated together and that nothing broke when a filter returned no data. I

am happy that the finished workbook meets the brief: it uses a real UK dataset, it is properly documented, and it gives a non-technical audience a clear, interactive way to explore where the UK’s greenhouse gas emissions come from.